ML

Monday, 25 May 2026 9:48 AM

Machine Learning
- Supervised
  - Regression
  - Classification
- Unsupervised
  - Clustering

① Supervised ML models

↳ Both labels and values are available

→ Types of supervised models
↳ ① Regression
② Classification

② Unsupervised Models

↳ Only values are available but labels are not available

→ Clustering is an example of unsupervised model

① Regression Model

↳ Type of supervised ML model which is used for predicting continuous numeric value (target) on the basis of few attributes values. (features)

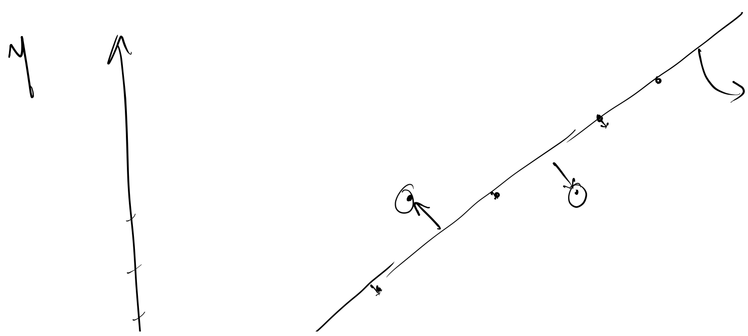
eg.

Height (x)	Weight (y)
5	6
6	8
7	10
8	12

⇒ Features (X) = ["Height"]

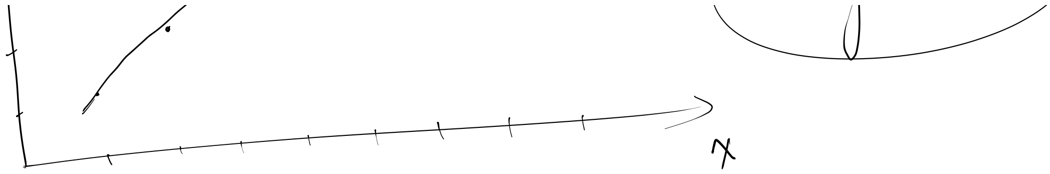
Target (Y) = ["weight"]

→ Identify the relation between X and Y.



↳ Best-Fit Line ⇒ The line which has minimum distance from all individual datapoints

$$y = mx + c$$



Practical Analysis

① Gather the required data from multiple sources.
- .csv
- .txt
- .db

② Convert all types of data into single usable format for data analysis

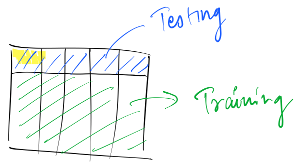
1-Dimensional data ⇒ Series
2-Dimensional data ⇒ DataFrame
} Pandas Datastructure

③ Perform data preprocessing to make our data ready for model building process. → data cleaning, data transformation, handling missing value, handling duplicate value

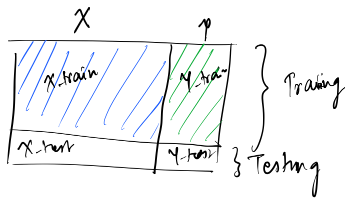
pandas < dropna(), fillna()

④ Define target(y) and features(x)

⑤ Split the dataset into training (90%) and testing (10%) set



⑥ Split the data into four parts
x_train, x_test, y_train, y_test
(train-test-split)



⑦ Model Selection is done
↳ model = LinearRegression()

⑧ Train the machine learning model with the help of training set of data. (x_train, y_train)
↳ model.fit(x_train, y_train)

⑨ Prediction can be done using our model by passing rest testing features (x_test).
y_pred = model.predict(x_test)

⑩ Compare predicted values (y_pred) to the actual value (y_test) to get the model accuracy.